

Day-3 Amplitudes: B01_ee_to_mumu

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Momentum Definitions

$$q_s = p_1 + p_2$$

Diagram Amplitudes

$$\mathcal{M}_s = \bar{v}(p_2)_{a_2} (-ie\gamma^{\mu_1}) u(p_1)_{a_1} \frac{-ig^{\mu_1\mu_2}}{q_s^2 + i\epsilon} \bar{u}(p_3)_{a_3} (-ie\gamma^{\mu_2}) v(p_4)_{a_4} \quad (1)$$

Total Amplitude

$$\mathcal{M} = \mathcal{M}_s \quad (2)$$

Rule/Source Audit

amplitude	rule id	source
amplitude:b01_ee_to_mumu	rules_tree_v1:ee-ee-gamma	docs/QED/GENERATION_AUDIT.md
amplitude:b01_ee_to_mumu	rules_tree_v1:mumu-gamma-gamma	docs/QED/GENERATION_AUDIT.md
amplitude:b01_ee_to_mumu	rules_tree_v1:photons-propagate	docs/QED/GENERATION_AUDIT.md